

Assignment 1

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1 Introduction

The aim of this assignment is to investigate the community structure of a network using different community detection algorithms.

The analyzed network represents similarities between anime extracted from the MyAnimeList dataset. Nodes represent anime, while edges represent similarity relationships based on shared genres. The dataset was filtered by considering only popular anime with at least 50,000 members and valid genre information.

Three community detection algorithms were applied: Louvain, Leiden, and Girvan–Newman. The obtained partitions were then compared using Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI) to evaluate the consistency between the detected communities.

2 Network Description

The graph was constructed from the filtered anime dataset. Each node represents an anime identified by its unique identifier. An undirected edge is created between two anime when they share at least two common genres.

The resulting network contains:

Property	Value
Number of nodes	3579
Number of edges	780393
Graph type	Undirected
Edge criterion	At least two shared genres

Table 1: Main properties of the anime similarity network.

Figure 1 shows the visualization of the anime similarity network.

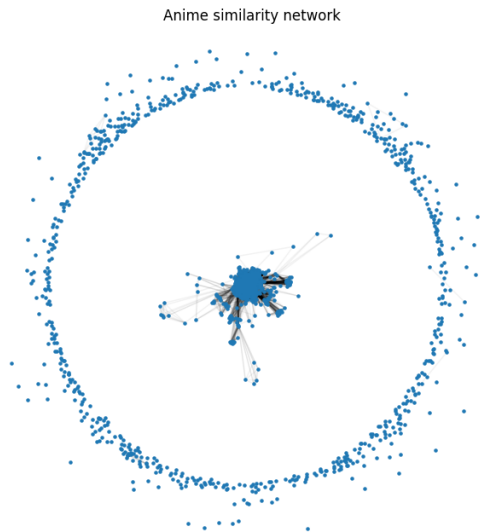


Figure 1: Visualization of the anime similarity network.

3 Community Detection

3.1 Louvain Algorithm

The Louvain algorithm was applied to the complete network. It detects communities by optimizing the modularity function, identifying groups of nodes with dense internal connections.

The algorithm detected 733 communities with an execution time of 5.380 seconds.

3.2 Leiden Algorithm

The Leiden algorithm was applied to the complete graph. It improves the Louvain approach by providing better connected communities while maintaining modularity optimization.

The algorithm detected 733 communities with an execution time of 0.389 seconds.

3.3 Girvan–Newman Algorithm

The Girvan–Newman algorithm detects communities by progressively removing edges with high betweenness centrality.

Due to its computational complexity, the algorithm was applied only to a reduced graph composed of 200 nodes.

The algorithm detected 48 communities with an execution time of 1.485 seconds.

Algorithm	Communities	Execution Time (s)
Louvain	733	5.380
Leiden	733	0.389
Girvan–Newman	48	1.485

Table 2: Comparison of community detection algorithms.

The results show that Louvain and Leiden produced the same number of communities on the complete graph. However, the Leiden algorithm required significantly less execution time, showing better computational efficiency. Girvan–Newman was applied to a reduced subgraph. Therefore, the number of detected communities cannot be directly compared with those obtained on the complete graph.

4 Evaluation of Community Quality

Since no ground truth community assignment is available, the detected partitions were compared using Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI). These metrics measure the similarity between different community assignments, without determining which partition is the correct one.

The comparison was performed between Louvain and Leiden, since both algorithms were applied to the complete graph.

Comparison	NMI	ARI
Louvain vs Leiden	0.9296	0.8763

Table 3: Similarity between community partitions.

The obtained values (NMI = 0.9296 and ARI = 0.8763) indicate a strong agreement between the Louvain and Leiden algorithms. Although the detected partitions are not identical, both algorithms identify a very similar underlying community structure. Girvan–Newman was not included in this comparison because it was applied only to a reduced

graph due to computational limitations.

5 Community Visualization

The detected communities were visualized by assigning different colors to the nodes according to their community label.

Figures 2, 3 and 4 show the partitions obtained with Louvain, Leiden and Girvan–Newman respectively.

5.1 Louvain Communities

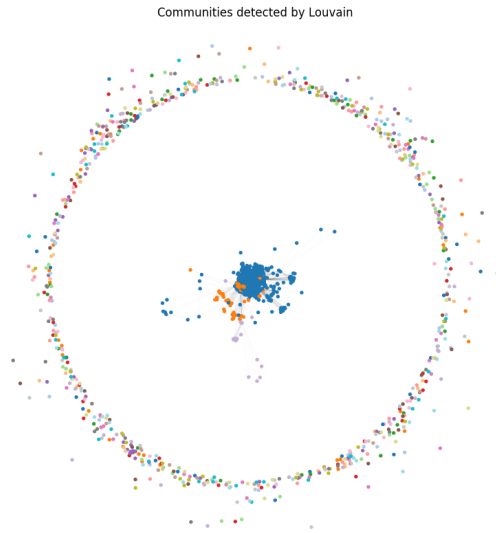


Figure 2: Communities detected by the Louvain algorithm.

5.2 Leiden Communities

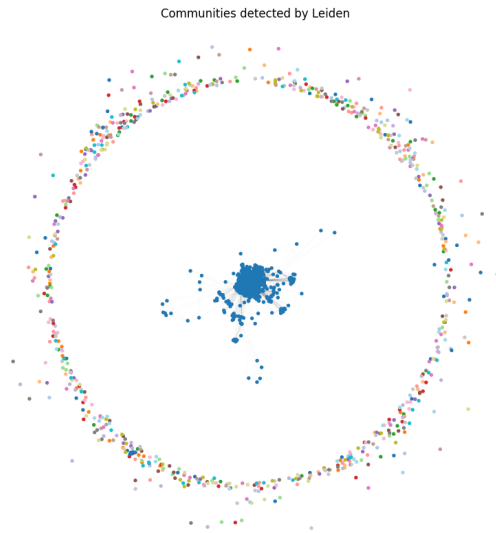


Figure 3: Communities detected by the Leiden algorithm.

5.3 Girvan–Newman Communities

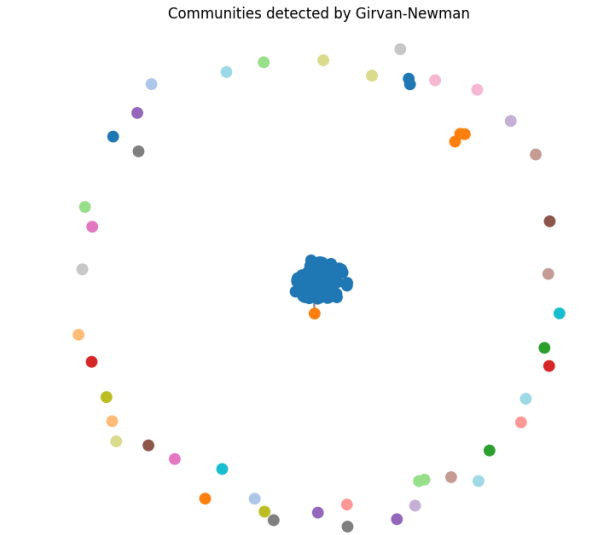


Figure 4: Communities detected by Girvan–Newman on the reduced graph.

6 Conclusions

This work analyzed the community structure of an anime similarity network using three different community detection algorithms.

Louvain and Leiden successfully identified communities in the complete network and showed a high level of agreement according to NMI and ARI metrics. Girvan–Newman provided an additional comparison, although its application was limited to a smaller graph because of its computational cost.

Overall, the results show that both Louvain and Leiden are effective for detecting communities in large similarity networks. Leiden achieved comparable community quality while requiring significantly less execution time, making it the most efficient algorithm among those tested.